

Report of Analysis



Intertek India Private Limited
"F" wing Tex Centre,
26-A, Chandivali Farm Road,
Off. Saki Vihar Road, Andheri (East)
Mumbai – 400 072, India.
Tel : +91 22 42450100
www.intertek.com

Report No.: 002378		Report Date: 08/07/2024
Customer Name:	SAVITA OIL TECHNOLOGIES LIMITED	
Contact Person:	Girish A Morde	
Address:	66/67, Nariman Bhavan, Nariman Point Mumbai - 400 021 Maharashtra, India	
Sample submitted as:	TRANSOL SYNTH – 100 500ml x Plastic Bottle	
Description on label/Sample Bottle:	TRANSOL SYNTH – 100	

Sample Number	18071214	Date of Sample Recieved:	15/06/2024
Date of Job started:	15/06/2024	Date of Job Completed:	08/07/2024

Method: OECD 201 (Freshwater Alga and Cyanobacteria, Growth inhibition test)

A study was performed to assess the adverse effects of TRANSOL SYNTH.100 on the growth rate (= rate of increase in cell density with time) and the yield (= biomass at time t minus initial biomass) of the planktonic freshwater algal species *Scenedesmus abundans* (Green algae) over several generations.

Sample Description:

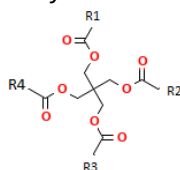
Product name : TRANSOL SYNTH.100

Type of product : Synthetic Ester based Dielectric fluid

CAS number : 68424-31-7

Sample Provided By: SAVITA OIL TECHNOLOGIES LIMITED

Structural Formulae: Pentaerythritol esters



Where,
R1, R2, R3 and R4 are alkyl groups may be same or different chain length.

Appearance and odor : Pale Yellow liquid

Physical state : Liquid

Vapor pressure (mm Hg) : <0.01 mm Hg at 200C

Solubility in water : <1 mg/l

Stability in light: Stable

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Test Specimen

Name: *Scenedesmus abundans* (Green algae)

Source: CSIR- National Chemical Laboratory (NCL)

Culture condition: Pre cultures were set up three days before the start of a test. They were grown under identical exposure conditions as the stock cultures

Test Conditions

Duration of Test: 72 Hours

Test Vessel: 250ml Erlenmeyer flasks with cotton stoppers

Culturing and Testing Media: OECD medium of OECD TG 201, annex 1)

Incubation Temperature: 21°C - 24°C

Light Intensity: 60-120 $\mu\text{E} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$

Test Volume: 100mL

pH of Test Solution: 7.9 - 8.1

Cells Density: 1.00×10^4 CFU/ml

Replicates: Six Replicates for each control and three replicates for each test concentration

Concentration Tested: 5.0mg/L, 10.0mg/L, 50mg/L, 100mg/L

Composition of Growth Medium: pH of the medium- 8.1

Nutrient	Final Concentration ($\mu\text{g/L}$)
NaNO ₃	25,500
MgCl ₂ •6H ₂ O	10,000
CaCl ₂ •2H ₂ O	4,420
H ₃ BO ₃	186
MnCl ₂ •4H ₂ O	416
ZnCl ₂	3.28
CoCl ₂ •6H ₂ O	1.43
CuCl ₂ •2H ₂ O	0.012
Na ₂ MoO ₄ •2H ₂ O	7.26
FeCl ₃ •6H ₂ O	160
Na ₂ EDTA•2H ₂ O	300
MgSO ₄ •7H ₂ O	14,700
K ₂ HPO ₄	1,044
NaHCO ₃	15,000

Apparatus

Analytical balance, pH meter, Shaking incubator, Microcell counter, Various glass materials: Erlenmeyer flasks, volumetric flasks, beakers, pipettes etc

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Preparation of pre cultures

Pre cultures were set up three days before the start of a test. They were grown under identical exposure conditions as the stock cultures, except from the use of a different growth medium (annex 1).

Preparation of the Test Solution

The test item was insoluble in water hence In order to maximize the amount of test item in solution, the test solutions were prepared in OECD media by adding 5.0mg, 10.0mg, 50mg, 100mg weight of the sample per liter of media, 10 mg of Polysorbate 80 used as a emulsifier for dissolution. The 100 mL of the solution was taken and 1 ml of the algal inoculum was added to each replicate resulting in a final cell density of 10000 cells/mL. All flasks were sealed with cotton stoppers

Test Results

Confirmatory Chemical Analysis

Measured Concentrations of TOC in Test Solutions

Nominal Loading Rate of Test Item (mg/L)	TOC (mg/L)			
	Initial	Final	Mean	SD
Control	<RDL	<RDL	NA	NA
5.0	<RDL	<RDL	NA	NA
10.0	<RDL	<RDL	NA	NA
50	<RDL	<RDL	NA	NA
100	<RDL	<RDL	NA	NA

RDL: Below reportable detection limit (5 mg/L)

pH in Test Solution

Nominal Loading Rate of Test Item (mg/L)	Initial pH (0 hour)	Final pH (72 hour)
Control	8.1	8.0
5.0	8.1	7.9
10.0	8.1	8.0
50	8.1	7.9
100	8.1	7.9

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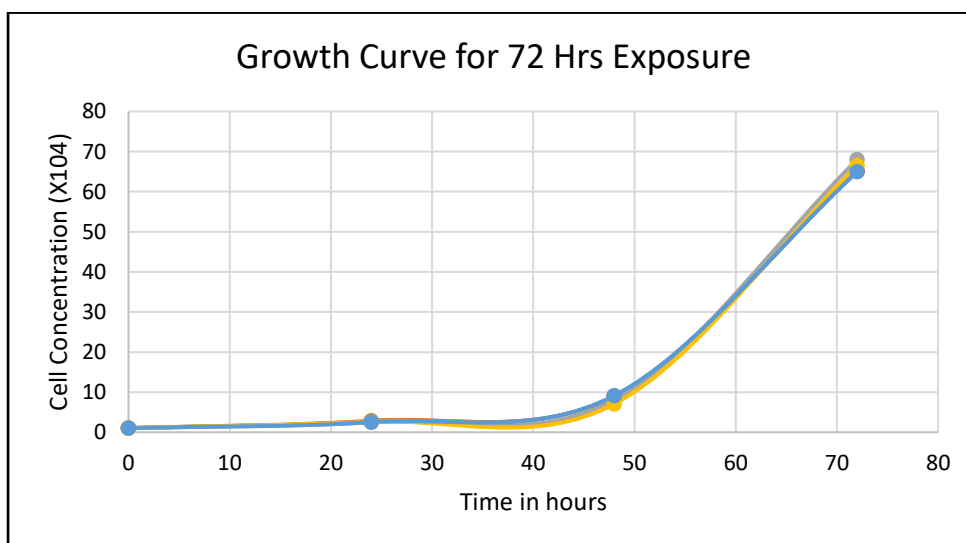
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Mean Cells Count in each 24 period

Nominal Loading Rate of Test Item (mg/L)	Mean Cell counts (x 10 ⁴ cells/ml)			
	0 hour	24 hours	48 hours	72 hours
Control	1.04	2.80	9.04	65.06
5.0	1.04	2.90	9.00	65.12
10.0	1.04	2.50	8.06	68.01
50	1.04	2.62	7.06	66.50
100	1.04	2.55	9.12	64.98



Graph 1: Growth Curve for 72 Hrs of Exposure for *Scenedesmus abundans*

Mean Specific Growth Rate over Time

Nominal Loading Rate of Test Item (mg/L)	Mean Specific Growth Rate over Time (day ⁻¹)
Control	0.89
5.0	0.89
10.0	0.93
50	0.90
100	0.88

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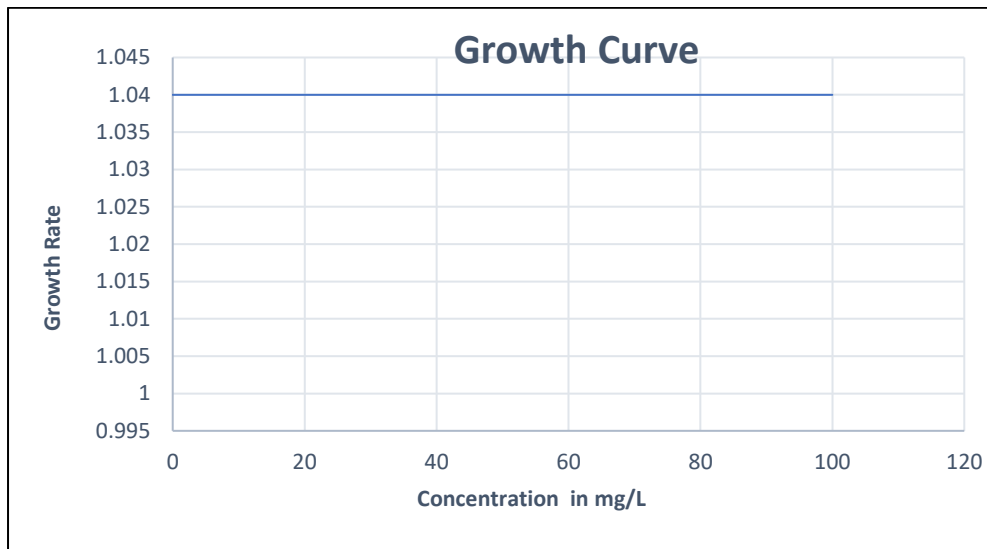
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Graph 2: Growth Rate Response curve of the effect

Summary of the Mean Inhibition % in cell yield and Specific growth rate

Nominal Loading Rate of Test Item (mg/L)	Mean Inhibition In yield (%)	Mean Inhibition In Specific Growth Rate (%)
Control	N/A	N/A
5.0	-1.2	0.00
10.0	2.0	-4.49
50	-3.4	-1.12
100	3.1	1.12

Summary of the Test Endpoint Values

Measurement Type	Cell Yield	Specific Growth Rate
E _r C 50 (mg/L)	>100	>100
E _r C 10 (mg/L)	>100	>100
NOEC (mg/L)	>100	>100
LOEC (mg/L)	>100	>100

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Conclusion

The given sample TRANSOL SYNTH.100 was tested to assess its toxic effects on the growth of *Scenedesmus abundans* (Green algae) as per OECD 201. No toxic effects against algae were observed at a limit test concentration of 100 mg/L.

This test was conducted using the following nominal loading rates: 5.0, 10, 50 and 100 mg test item/L in addition to a negative control.

Total organic carbon was analyzed as a proxy for the test item therefore, the biological test endpoints were calculated using the nominal loading rates of the test item.

The 72 hours ErC 50, ErC 10, and NOEC (mg/L) and LOEC (mg/L) for cell yield and specific growth rate are summarized in the table below, based on the nominal loading rate of

Measurement Type	Cell Yield	Specific Growth Rate
ErC 50 (mg/L)	>100	>100
ErC 10 (mg/L)	>100	>100
NOEC (mg/L)	>100	>100
LOEC (mg/L)	>100	>100

Authorized Signatory



Akshata Rangle
Assistant Manager - Microbiology

----- End of Report -----

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